

Group 4: Vitamin D Supplementation Study

Dataset Guidance

Dataset Overview

File: vitamin_d.csv

Sample Size: $n = 45$ adults (15 per dose group)

Research Context: Randomized controlled trial comparing three vitamin D supplement doses (Low, Medium, High) on vitamin D blood levels after 8 weeks.

Variables

- **id:** Participant identifier (1-45)
- **group:** Supplement dose group (Low, Medium, High)
- **baseline:** Baseline vitamin D level before supplementation (ng/mL)
- **vitamin_d:** Vitamin D level after 8 weeks of supplementation (ng/mL)
- **age:** Age in years

Loading the Data in R

```
# Load the dataset
data <- read.csv("vitamin_d.csv")

# View structure
str(data)
head(data)

# Summary by group
tapply(data$vitamin_d, data$group, summary)
```

Research Question to Consider

“Do different vitamin D supplement doses lead to different vitamin D blood levels?”

Alternatively: “Which supplement dose is most effective?”

Key Decision Point

The Challenge: You have baseline AND final measurements for each person. How will you use the baseline information in your analysis using methods covered in this course?

Questions to explore with AI:

- What are different ways to handle baseline measurements?
- Should you analyze final levels? Change scores? Both?
- What methods from this course can compare three groups?

Important note: AI may suggest approaches for handling baseline that you haven't learned yet. Consider what's possible with methods covered in this course.

Things to Think About

Exploratory Analysis:

- Create visualizations (boxplots by group for final levels, baseline, or change)
- Calculate summary statistics by supplement dose group
- Consider: What question are you trying to answer about the doses?

Statistical Methods:

- Review methods for comparing groups
- Consider: How many groups are you comparing?
- Consider: What outcome should you analyze?

Baseline Measurements:

- You have baseline vitamin D levels - how could you use this information?
- Different approaches: ignore baseline, calculate change, adjust for baseline
- Which approaches are possible with methods you've learned?

AI Consultation Strategy

Important: When you ask AI about studies with baseline measurements, it may suggest advanced methods beyond this course.

Prompts to try:

- “How should I analyze this vitamin D supplement study?”
- “What’s the best way to handle baseline measurements in this analysis?”
- “I haven’t learned [method AI suggested]. Can I answer my question with ANOVA?”
- “What’s the difference between analyzing final levels vs. change scores?”

Watch for:

- Tools suggesting different approaches to baseline measurements
- Methods you haven’t encountered in this course
- Trade-offs between simple and complex approaches

Document everything: Record what each AI tool recommends - this is material for your AI comparison section!

Questions for Your Analysis

As you work through this dataset, consider:

1. What research question are you answering?
2. Which statistical method(s) from this course can address this question?
3. How are you handling the baseline measurements? Why?
4. What assumptions do you need to check?
5. If AI suggested advanced methods for baseline adjustment, why did you choose your approach?

Possible Approaches to Explore

Think about these options (all use methods from this course):

- **Option 1:** Analyze final vitamin D levels only
- **Option 2:** Calculate change scores and analyze those
- **Option 3:** Compare both approaches

Which makes most sense for your research question? Why?

Resources

- Week 9 drop-in session (Wed 12-1:20pm, 153 Milam)
- Textbook: Chapter 12 (Analysis of Variance)
- Your team!
- Canvas discussion board

Remember: Multiple valid approaches may exist. The goal is to:

- Choose a method you understand from this course
- Justify why your approach answers the research question
- Explain your reasoning about baseline measurements
- Critically evaluate what different AI tools suggested
- Document your decision-making process in your report